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The Influence of Detailed Chemical Kinetics and Radiation on the Post-Detonation Afterburning of TNT

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Numerical simulations were performed in 3D Cartesian coordinates to explore the post-detonation afterburning processes following the detonation of hemispherical TNT explosive charges. The simulations captured air dissociation behind the primary shock, chemical equilibrium within the early-time fireball, and afterburning between the detonation products and air. The gas-phase uses the Becker-Kistiakowsky-Wilson equation of state. Finite-rate chemical kinetics are considered using a skeletal chemical mechanism of 43 species and 268 reactions. The afterburning processes are coupled to the detonation with a simplified programmed burn approach. Five simulations were performed with three different charge diameters of 12, 120, and 1200 mm. Three simulations included particle thermal radiation and the smallest and larges charges were rerun as non-radiative cases. The simulations are nearly self-similar at early time, yet discrepancies in mixing and chemical timescales due to charge size result in differences in late time fireball temperature and chemical composition. A Damköhler analysis shows that the chemical reactions are mixing-limited in the mixing layer of all three charge sizes. Thermal particle radiation is shown to have little impact, especially in the smallest charge.

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I. Introduction

The afterburning of fuel-rich product gases following the detonation of an explosive charge releases substantial chemical energy [1]. These afterburning processes act on much longer timescales than the initial detonation [2] and can be heavily influenced by scale, chemical composition, and the existence of particles [3].

Detonations of explosive charges are often divided into five stages [4]. In stage 1, the detonation propagates through the charge and converts the solid explosive into hot detonation products.

Stage 2 occurs when the detonation breaks out of the charge. A primary shock propagates into the air, dissociating it and raising local temperatures in excess of 12,000 K. An expansion wave propagates back towards the origin, expanding the flow, and generating a return shock. A contact surface forms between the detonation products and shock-heated air where density gradients between the two result in the formation of Rayleigh-Taylor instabilities [5]. The instabilities grow, mixing the detonation products and the air as the flow continues to expand, promoting chemical reactions.

The return shock initially moves outward but is an inward running shock and eventually reverses direction as the expansion of the detonation products slows. This begins stage three. The return shock moves back towards the origin, further entraining air into the fireball. Stage four begins when the return shock rebounds off the origin, creating the reshock. The reshock moves through the fireball, generating Richtmyer-Meshkov [6, 7] instabilities, greatly increasing mixing. Stage 5 is represented by continued mixing and late-stage afterburning of the remaining detonation products.

Many studies have used simplifications to model the chemical kinetics, such as global single-step mechanisms and the infinite-rate chemistry assumption [8–12]. Our prior work explored the importance of finite-rate detailed chemical kinetics on the post-detonation afterburning of pentaerythritol tetranitrate (PETN) [13, 14]. PETN has a low oxygen balance (-10%), produces very little soot, and shows signs of more mild afterburning. Explosives with more negative oxygen balances, such as Trinitrotoluene (TNT) (-74%), produce detonation products with up to 20% carbon soot particles by mass. These fuel-rich detonation products can produce much more substantive afterburning. Carbon soot particles can react with gas-phase species, sublimate into gas-phase carbon, and agglomerate. A mechanism for the carbon soot particles was developed to include these physics. Prior studies have also shown that particles can impact the post-detonation mixing of the fireball [3]. Therefore, it is important to include particles.

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Thermal radiation is often neglected in the simulation of high explosive blasts. Radiation timescales are considered relatively slow, and the impact on the flow is considered negligible. Modeling the radiation also adds an additional layer of computational complexity. However, radiation has been shown to influence dust explosions [15–18]. Due to the high mass of carbon soot particles present in the detonation products of TNT, we chose to consider thermal radiation from the particles.

The current work looks to explore the impact of finite-rate detailed chemical kinetics and carbon soot particle burning and radiation on the post-detonation afterburning of TNT. TNT was chosen because of its highly negative oxygen balance (-74%) and fuel-rich detonation products. Additionally, the impact of charge size is explored through a parametric study in both radiative and non-radiative cases.

II. Simulation Setup

A. Geometrical Setup

Center-initiated hemispherical TNT charges were detonated in air at 298 K and 1 bar. The initial density of the charges was taken to be 1654 kg/m^3. The detonation velocity, enthalpy of formation, and specific heat of TNT were taken to be 7090 m/s [19, 20], -277.605 KJ/kg [19, 20], and 1143 J/kg·K [21]. The detonation product mass fractions at the Chapman-Jouguet (CJ) state were calculated with the Tiger equilibrium code for pure TNT at 1654 kg/m^3 using the JCZ3 library [19, 20]. The charge diameter, domain boundaries, and simulation time were varied by a scaling factor, f. Five cases are considered. The simulation setups are presented in Table 1. The geometrical setup in 2D is shown in Fig. 1

Table 1 Simulation Setup for Hemispherical TNT Charges

Case	f	Charge Diameter	Domain	End Time	Radiation
1	1	12 mm	0.3 m x 0.3 m x 0.3 m	350 μs	Yes
2	10	120 mm	3 m x 3 m x 3 m	3500 μs	Yes
3	100	1200 mm	30 m x 30 m x 30 m	35000 μs	Yes
4	1	12 mm	0.3 m x 0.3 m x 0.3 m	350 μs	No
5	100	1200 mm	30 m x 30 m x 30 m	35000 μs	No

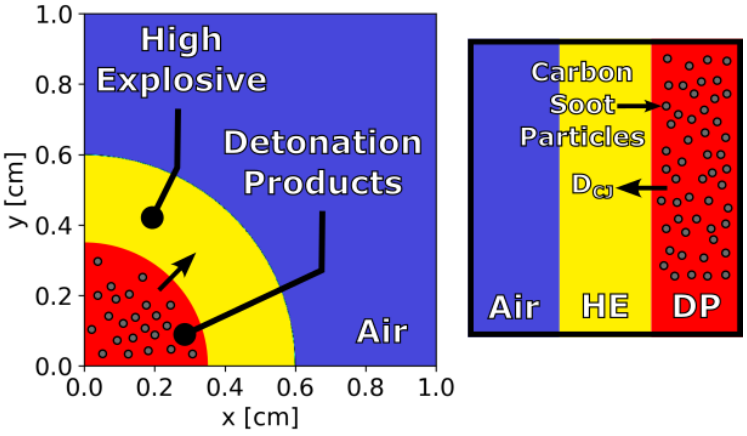


Fig. 1 An example geometrical configuration for the numerical simulation as the detonation propagates through a 12-mm diameter TNT charge.

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B. Governing Equations

The simulations solve the reactive Euler equations for a three-phase mixture of a reactive gas (subscript g), reactive particles (subscript s), and a solid explosive (subscript h). The mixture is split between a continuous phase consisting of the gas-phase and solid explosive and a dispersed phase consisting of particles. The system uses the five-equation multiphase flow model [22–24] for the continuous phase combined to a Eulerian granular model [25] for the dispersed phase. The volume fractions for the continuous and dispersed are related through

$$\alpha_\mathrm{c} + \alpha_\mathrm{s} = 1 \tag{1}$$

where α_c and α_s are the volume fractions of the continuous and dispersed phase, respectively. The continuous phase volume fractions are related through

$$\alpha_\mathrm{g} + \alpha_\mathrm{h} = 1 \tag{2}$$

where α_h is the volume fraction for the solid-phase explosive.

The continuous phase is assumed to be in thermal and mechanical equilibrium

$$\begin{aligned} &1)\ \ \mathbf{v}_\mathrm{g} = \mathbf{v}_\mathrm{h} \\ &2)\ \ p_\mathrm{g} = p_\mathrm{h} \\ &3)\ \ T_\mathrm{g} = T_\mathrm{h} \end{aligned}$$

where $\mathbf{v}_\mathrm{g}$, p_g, and T_g are the velocity vector, pressure, and temperature of the gas-phase while $\mathbf{v}_\mathrm{h}$, p_h, and T_h are the velocity vector, pressure, and temperature of the solid-phase explosive. The assumptions reduced the governing equations to

$$\frac{\partial \alpha_\mathrm{h}}{\partial t} + \nabla \cdot (\alpha_\mathrm{h} \mathbf{v}_\mathrm{c}) = \alpha_\mathrm{h} \nabla \cdot \mathbf{v}_\mathrm{c} - \dot{\alpha}, \tag{3}$$

$$\frac{\partial \alpha_\mathrm{c} \alpha_\mathrm{h} \rho_\mathrm{h}}{\partial t} + \nabla \cdot (\alpha_\mathrm{c} \alpha_\mathrm{h} \rho_\mathrm{h} \mathbf{v}_\mathrm{c}) = -\mathbf{S}_\mathrm{h}^M \tag{4}$$

$$\frac{\partial \alpha_\mathrm{c} \alpha_\mathrm{g} \rho_\mathrm{g}}{\partial t} + \nabla \cdot \left(\alpha_\mathrm{c} \alpha_\mathrm{g} \rho_\mathrm{g} \mathbf{v}_\mathrm{c}\right) = \mathbf{S}_\mathrm{s}^M + \mathbf{S}_\mathrm{h}^M, \tag{5}$$

$$\frac{\partial \alpha_\mathrm{c} \alpha_\mathrm{g} \rho_\mathrm{g} Y_{\mathrm{g},i}}{\partial t} + \nabla \cdot \left(\alpha_\mathrm{c} \alpha_\mathrm{g} \rho_\mathrm{g} Y_{\mathrm{g},i} \mathbf{v}_\mathrm{c}\right) = \alpha_\mathrm{c} \alpha_\mathrm{g} \dot{\omega}_{\mathrm{g},i} + \mathbf{S}_{\mathrm{g},i}^Y + \mathbf{S}_{\mathrm{g},i}^\mathrm{h}, \tag{6}$$

$$\frac{\partial \alpha_\mathrm{c} \rho_\mathrm{c} \mathbf{v}_\mathrm{c}}{\partial t} + \nabla \cdot (\alpha_\mathrm{c} \rho_\mathrm{c} \mathbf{v}_\mathrm{c} \mathbf{v}_\mathrm{c}) + \nabla \alpha_\mathrm{c} p_\mathrm{c} = p_\mathrm{c} \nabla \alpha_\mathrm{c} + \mathbf{S}^p, \tag{7}$$

$$\frac{\partial \alpha_\mathrm{c} \rho_\mathrm{c} E_\mathrm{c}}{\partial t} + \nabla \cdot \left[\alpha_\mathrm{c} \mathbf{v}_\mathrm{c} \left(\rho_\mathrm{c} E_\mathrm{c} + p_\mathrm{c}\right)\right] = -p_\mathrm{c} \nabla \cdot \alpha_\mathrm{s} \mathbf{v}_\mathrm{s} + \mathbf{S}_\mathrm{c}^E, \tag{8}$$

where ρ_g, $Y_{\mathrm{g},i}$, and $\dot{\omega}_{\mathrm{g},i}$ are the density, mass fraction of gas-phase species species i, and homogeneous mass rate of production for gas-phase species i, respectively. ρ_h is the density of the solid-phase explosive. $\dot{\alpha}$ is the rate of volume change due to depletion of solid-phase explosive. ρ_c, p_c and E_c are the density, pressure, and total energy for the continuous phase, respectively. α_s and $\mathbf{v}_\mathrm{s}$ are the volume fraction and velocity vector of the particle-phase. The interface coupling terms for mass, momentum, and energy between the gas, particle, and solid explosive are $\mathbf{S}_p^M$, $\mathbf{S}_\mathrm{h}^M$, $\mathbf{S}_{\mathrm{g},i}^Y$, $\mathbf{S}_{\mathrm{g},i}^\mathrm{h}$, $\mathbf{S}^p$, and $\mathbf{S}_\mathrm{c}^E$.

The total energy of the gas-phase is

$$E_\mathrm{g} = e_\mathrm{g} + \frac{1}{2} \mathbf{v}_\mathrm{g} \cdot \mathbf{v}_\mathrm{g}, \tag{9}$$

where e_g is the internal energy of the gas-phase. The total energy of the solid-phase explosive is

$$E_\mathrm{h} = e_\mathrm{h} + \frac{1}{2} \mathbf{v}_\mathrm{h} \cdot \mathbf{v}_\mathrm{h}, \tag{10}$$

where e_h is the internal energy of the solid-phase explosive.

The continuous phase mixture mass and energy is taken to be

$$\rho_\mathrm{c} = \alpha_\mathrm{g} \rho_\mathrm{g} + \alpha_\mathrm{h} \rho_\mathrm{h}, \tag{11}$$

$$\rho_\mathrm{c} e_\mathrm{c} = \alpha_\mathrm{g} \rho_\mathrm{g} e_\mathrm{g} + \alpha_\mathrm{h} \rho_\mathrm{h} e_\mathrm{h}. \tag{12}$$

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The dispersed phase makes the assumption of spherical, monodisperse particle. The governing equations are

$$\frac{\partial \alpha_\mathrm{s}\rho_\mathrm{s}}{\partial t} + \nabla\cdot(\alpha_\mathrm{s}\rho_\mathrm{s}\mathbf{v}_\mathrm{s}) = -\mathbf{S}_\mathrm{s}^M + \mathbf{S}_\mathrm{h}^M, \tag{13}$$

$$\frac{\partial \alpha_\mathrm{s}\rho_\mathrm{s}\mathbf{v}_\mathrm{s}}{\partial t} + \nabla\cdot(\alpha_\mathrm{s}\rho_\mathrm{s}\mathbf{v}_\mathrm{s}\mathbf{v}_\mathrm{s}) + \nabla p_\mathrm{s} + \nabla p_{fric} = -\alpha_\mathrm{s}\nabla p_\mathrm{c} - \mathbf{S}^p, \tag{14}$$

$$\frac{\partial \alpha_\mathrm{s}\rho_\mathrm{s}E_\mathrm{s}}{\partial t} + \nabla\cdot(\alpha_\mathrm{s}\rho_\mathrm{s}E_\mathrm{s}\mathbf{v}_\mathrm{s}) = -p_\mathrm{s}\nabla\cdot\mathbf{v}_\mathrm{s} + \mathbf{S}_\mathrm{s}^{PTE}, \tag{15}$$

$$\frac{\partial \alpha_\mathrm{s}\rho_\mathrm{s}e_\mathrm{s}}{\partial t} + \nabla\cdot(\alpha_\mathrm{s}\rho_\mathrm{s}e_\mathrm{s}\mathbf{v}_\mathrm{s}) = \mathbf{S}_\mathrm{s}^E + \mathbf{S}_\mathrm{s}^{rad} \tag{16}$$

where ρ_s, p_s, p_{fric}, E_s, and e_s are the density, solids pressure from kinetic theory, frictional-collisional pressure, pseudo-thermal energy (PTE), and internal energy of the particle-phase, respectively. The interphase coupling terms $\mathbf{S}_\mathrm{s}^{PTE}$ and $\mathbf{S}_\mathrm{s}^E$ are expanded upon in [25] and $\mathbf{S}_\mathrm{s}^{rad}$ is discussed later. ρ_s is taken to be constant. The PTE, $E_\mathrm{s} = \frac{3}{2}\Theta_\mathrm{s}$, is the energy due to random translational motion of the particles and is a function of the granular temperature, Θ_s. The granular temperature is the mean square of the particle velocity fluctuations. The internal energy of the particle-phase is given by

$$e_\mathrm{s} = e_{\mathrm{s},f}^0 + C_{V,\mathrm{s}}\left(T_\mathrm{s} - T_\mathrm{s,ref}\right) \tag{17}$$

where $e_{\mathrm{s},f}^0$ is the internal energy of formation, $C_{V,\mathrm{s}}$ is the constant-volume specific heat of the particle-phase, T_s is the 'normal' particle-phase temperature, and $T_\mathrm{s,ref}$ is the particle-phase reference temperature. This implementation assumes a small Biot modulus from a uniform tempreature distribution within each particle. p_s represents the collisional effects of particles. It comes from an equation of state for a granular gas

$$p_\mathrm{s} = \alpha_\mathrm{s}\rho_\mathrm{s}\Theta_\mathrm{s}\left[1 + 2(1+e)\alpha_\mathrm{s}g_0\right] \tag{18}$$

where e is the coefficient of restitution and g_0 is the radial distribution function. g_0 is given as

$$\frac{1}{g_0} = 1 - \left(\frac{\alpha_\mathrm{s}}{\alpha_{\mathrm{s},max}}\right)^{1/3}, \tag{19}$$

where $\alpha_{\mathrm{s},max}$ is the packing limit, set for a monodisperse particle phase to 0.65 in this work. The frictional-collisional pressures is given as

$$p_{fric} = \begin{cases} 0 & \text{if } \alpha_\mathrm{s} < \alpha_{\mathrm{s},crit}, \\ 0.1\alpha_\mathrm{s}\frac{\left(\alpha_\mathrm{s}-\alpha_\mathrm{s,crit}\right)^2}{\left(\alpha_\mathrm{s,max}-\alpha_\mathrm{s}\right)^5} & \text{if } \alpha_\mathrm{s} \geq \alpha_{\mathrm{s},crit}, \end{cases} \tag{20}$$

where $\alpha_{\mathrm{s},crit}$ is taken to be 0.5.

The radiation transport equation (RTE) is solved to account for thermal radiation from the particles. The RTE is

$$\frac{1}{c}\frac{\partial I}{\partial t} + \Omega\cdot\nabla I + \sigma_s I = \frac{1}{4\pi}\int_{4\pi}\sigma_s I\left(r,\Omega'\right)d\Omega' + \sigma_{a,\mathrm{s}}\left(\frac{acT_\mathrm{s}^4}{2\sqrt{\pi}} - I\right) \tag{21}$$

where c is the speed of light, I is the radiation intensity, Ω is the angular variable, σ_s is the scattering coefficient, $\sigma_{a,\mathrm{s}}$ is the absorption coefficient for the particle-phase, and a is a radiation constant given as 1.372E13 $\mathrm{Jm^{-3}KeV^{-4}}$. Incident radiation, G, is defined as

$$G = \int_{4\pi} I d\Omega. \tag{22}$$

The RTE is simplified using third-order filtered spherical harmonics ($\mathrm{FP_3}$) [26] using the two assumptions of isotropic scattering and that the radiation is spectrally gray. We take the gas-phase to be optically thin and therefore noncontributory to radiation. The particle phase uses an average of the Planck-mean and Rosseland-mean extinction and absorption coefficients [27],

$$\kappa_\mathrm{s} = \beta_\mathrm{s} = 3.72\alpha_\mathrm{s}C_0T_\mathrm{s}/C_2, \tag{23}$$

where C_2 is the second Planck function constant with a value of 1.4388 cm K and C_0 is

$$C_0 = \frac{36\pi nk}{\left(n^2-k^2+2\right)^2 + 4n^2k^2}, \tag{24}$$

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where n is the refractive index and k is the absorptive index from the complex index of refraction $m = n - ik$ [28].

The radiation heat source term is

$$\mathbf{S}_{\mathrm{s}}^{rad} = -\nabla \cdot q_{rad,\mathrm{s}} = \sigma_{a,\mathrm{s}} \left(G - 4\pi I_{b,\mathrm{s}} \right), \tag{25}$$

where $q_{rad,\mathrm{s}}$ and $I_{b,\mathrm{s}}$ are the radiation heat flux and blackbody intensity for the particles, respectively.

C. Equations of State and Thermodynamic Data

The Becker-Kistiakowsky-Wilson (BKW) equation of state (EOS) was used for the gas-phase [29–31]. BKW is a covolume EOS that is thermodynamically complete and works well with reactive mixtures, is applicable at explosive detonation conditions, and limits to ideal-gas at low pressures. The BKW EOS is

$$p_{\mathrm{g}} = \rho_{\mathrm{g}} R T_{\mathrm{g}} Z,\ Z = 1 + x e^{\beta x},\ x = \frac{\rho \sum K_i Y_{\mathrm{g},i}}{(T+\theta)^{\alpha}},\ \text{and } K_i = \frac{\kappa \hat{K}_i}{MW_i}, \tag{26}$$

where p_{g} is the gas-phase pressure, ρ_{g} is the gas-phase density, R is the gas constant, T_{g} is gas-phase temperature, Z is the compressibility factor, $Y_{\mathrm{g},i}$ is the mass fraction of gas-phase species i, $\hat{K}_i$ is the molar covolume factor, and MW_i is the molecular weight of gas-phase species i. β, θ, α, and κ are the BKW parameters [30] with values of 0.3, 6620, 0.5, and 10.5, respectively. The $\hat{K}_i$ are taken from the database provided by Hobbs and Baer [29]. $\hat{K}_i$ for species not found in the database are calculated using the Open babel code [32] following the same procedures described in [29]. Comparing values between both sources show agreement within a few percent

The internal energy for the gas-phase comes from the departure function [33]

$$\frac{e_{\mathrm{g}} - e_{\mathrm{g}}^{ig}}{RT_{\mathrm{g}}} = -\int_0^{\rho_{\mathrm{g}}} \frac{T_{\mathrm{g}}}{\rho_{\mathrm{g}}} \left(\frac{\partial Z}{\partial T_{\mathrm{g}}} \right)_{\rho_{\mathrm{g}}} d\rho_{\mathrm{g}} = \frac{\alpha T_{\mathrm{g}} x}{T_{\mathrm{g}} + \theta} e^{\beta x}, \tag{27}$$

where e_{g}^{ig} is the internal energy of an ideal-gas defined by

$$e_{\mathrm{g}}^{ig}(T_{\mathrm{g}}) = \int_{T_{\mathrm{g,ref}}}^{T_{\mathrm{g}}} C_{V,\mathrm{g}}^{ig}(\theta) d\theta + C_1, \tag{28}$$

where $C_{V,\mathrm{g}}^{ig}$ is the specific heat at constant volume for an ideal-gas and C_1 is a constant.

The gas-phase thermodynamic data is taken from the NASA9 polynomials. The polynomials are valid to 20,000 K for many species [34]. The temperatures following the detonation of an explosive blast can range above 12,000 K and well below ambient. We assume constant specific heats above and blow the limits of the NASA9 polynomials where data can not be found. The polynomials for the gas-phase specific heat and internal energy are

$$C_{V,\mathrm{g}}^{ig} = \sum_{i=1}^{N_{\mathrm{g}}} Y_{\mathrm{g},i} C_{V,\mathrm{g},i}^{ig}, \tag{29}$$

$$e_{\mathrm{g}}^{ig} = \sum_{i=1}^{N_{\mathrm{g}}} Y_{\mathrm{g},i} e_{\mathrm{g},i}^{ig}, \tag{30}$$

where for each gas-phase species

$$C_{V,\mathrm{g},i}^{ig} = \frac{Ru}{MW_i} \left[\frac{a_{1,i}}{T_{\mathrm{g}}^2} + \frac{a_{2,i}}{T_{\mathrm{g}}} + (a_{3,i} - 1) + a_{4,i} T_{\mathrm{g}} + a_{5,i} T_{\mathrm{g}}^2 + a_{6,i} T_{\mathrm{g}}^3 + a_{7,i} T_{\mathrm{g}}^4 \right], \tag{31}$$

$$e_{\mathrm{g},i}^{ig} = \frac{Ru}{MW_i} \left[a_{1,i} \ln(T_{\mathrm{g}}) - \frac{a_{2,i}}{T_{\mathrm{g}}} + (a_{3,i} - 1) T_{\mathrm{g}} + \frac{1}{2} a_{4,i} T_{\mathrm{g}}^2 + \frac{1}{3} a_{5,i} T_{\mathrm{g}}^3 + \frac{1}{4} a_{6,i} T_{\mathrm{g}}^4 + \frac{1}{5} a_{7,i} T_{\mathrm{g}}^5 + a_{8,i} \right]. \tag{32}$$

$a_{1,i}$, $a_{2,i}$, $a_{3,i}$, $a_{4,i}$, $a_{5,i}$, $a_{6,i}$, $a_{7,i}$, and $a_{8,i}$ are the polynomial coefficients for species i.

A barotropic EOS [24] is used for the solid explosive. The equation is

$$p_{\mathrm{h}} - p_{\mathrm{h},0} = a^2 D_{\mathrm{CJ}}^2 \left(\rho_{\mathrm{h}} - \rho_{\mathrm{h},0} \right), \tag{33}$$

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where $p_{\mathrm{h},0}$ is the reference pressure for the solid explosive, $a = 0.707$, D_{CJ} is the detonation velocity, and $\rho_{\mathrm{h},0}$ is the initial density of the explosive.

Non-ideal compression terms are neglected in the the internal energy for the high explosive [35].The equation is

$$e_{\mathrm{h}} = h^0_{f,\mathrm{h}} + C_{V,\mathrm{h}}(T_{\mathrm{h}} - T_{\mathrm{h,ref}}), \tag{34}$$

where $h^0_{f,\mathrm{h}}$ is the heat of formation of the explosive, $C_{V,\mathrm{h}}$ is the specific heat at constant volume of the explosive, T_{h} is the temperature of the explosive, and $T_{\mathrm{h,ref}}$ is 298.15 K.

A pressure-temperature equilibration procedure is used to find the mixture pressure and temperature by solving Eqns. 11 and 12.

D. Reaction Models

The gas-phase chemistry was evolved with a skeletal chemical mechanism generated from a detailed chemical reaction mechanism originally developed for propellants [36]. The skeletal chemical mechanism includes 43 species and 268 reactions. The gas-phase homogeneous reaction rate is

$$\dot{\omega}_{\mathrm{g},i} = \sum_j \dot{\omega}_{\mathrm{g},i,j}, \tag{35}$$

where the production rate of species i due to reaction j is

$$\dot{\omega}_{\mathrm{g},i,j} = MW_i \sum_j (\nu''_i - \nu'_i) \times \left[k_{f,j} \prod_{i=1}^{m} (C_i)^{\nu'_i} - k_{r,j} \prod_{i=1}^{m} (C_i)^{\nu''_i} \right]. \tag{36}$$

C_i, $k_{f,j}$, $k_{r,j}$, ν'_i, and ν''_i are the molar concentration of species i, forward and reverse rate constants for reaction j and stoichiometric coefficients of the reactants and products for species i in reaction j, respectively.

A simplified burn model (SBM) is used to convert the solid-phase explosive to the gas- and particle-phases [24]. The solid-phase explosive is converted into detonation products at the CJ state given by the Tiger equilibrium code using the JCZ3 library [19, 20]. The mass exchange due to the solid-phase explosive reactions is

$$\mathbf{S}^M_{\mathrm{h}} = \dot{M}_{\mathrm{h}} = \rho_{\mathrm{h}} \dot{\alpha}, \tag{37}$$

$$\dot{M}_{\mathrm{h},i} = \dot{M}_{\mathrm{h}} Y^{\mathrm{CJ}}_{\mathrm{g},i}, \tag{38}$$

$$\dot{M}_{\mathrm{h,s}} = \dot{M}_{\mathrm{h}} Y^{\mathrm{CJ}}_{\mathrm{s}}, \tag{39}$$

where $\dot{M}_{\mathrm{h}}$ and $\dot{M}_{\mathrm{h,s}}$ are the mass exchange for gas-phase species i and the particle-phase due to solid-phase explosive burning, respectively. $Y^{\mathrm{CJ}}_{\mathrm{g},i}$ and $Y^{\mathrm{CJ}}_{\mathrm{s}}$ are the mass fraction of gas-phase species i and the particle-phase at the Chapman-Jouguet (CJ) state, respectively. The mass fractions of the gas-phase species and particle-phase at the CJ state are related by

$$Y^{\mathrm{CJ}}_{\mathrm{s}} + \sum_i^{N_{\mathrm{g}}} Y^{\mathrm{CJ}}_{\mathrm{g},i} = 1. \tag{40}$$

$\dot{\alpha}$ is the rate of volume change from the solid- to the gas- and particle-phases due to explosive burn. A time-based ignition criterion is used where the solid-phase explosive ignites based on a time of arrival. The time of arrival can be calculated using burn tables based on detonation shock dynamics [37] or Huygens propagation rules.

The particle-phase is evolved using a reaction mechanism developed for carbon soot. The mechanism considers carbon soot reactions (oxidation and gasification), formation, sublimation, and agglomeration. Four carbon soot reactions are considered. The reaction rates for each reaction are sourced from [38]. The oxidation reactions are

$$C(s) + O_2 \rightarrow CO_2, \tag{41}$$

$$C(s) + OH \rightarrow CO + H, \tag{42}$$

and the gasification reactions are

$$C(s) + CO_2 \rightarrow 2CO, \tag{43}$$

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$$C(s) + H_2O \rightarrow CO + H_2. \tag{44}$$

A simple soot formation model based on the precursor C_2H_2 is implemented [39, 40]. The model includes soot formation due to nucleation and surface growth. The nucleation reaction is

$$C_2H_2 \rightarrow 2C(s) + H_2, \tag{45}$$

and the surface growth reaction is

$$C_2H_2 + nC(s) \rightarrow (n+2)C(s) + H_2. \tag{46}$$

A model is implemented to combine kinetic- and diffusion-limited sublimation regimes using a hybrid resistor approach. Figure 2 shows a graphical representation of the model. The kinetic-limited mass flux is represented by the Schrage-Knudsen equation [41] where the mass flux is limited by the creation of gaseous carbon. The diffusion-limited mass flux is represented by Fick's law of diffusion [42] where the mass flux is limited by the transport of gaseous carbon away from the particle surface.

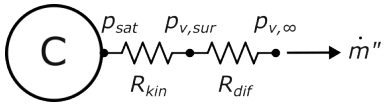


Fig. 2 A graphical representation of the hybrid resistor model used for sublimation. $p_{\mathbf{sat}}$, $p_{v,\mathbf{sur}}$, and $p_{v,\infty}$ are the saturation, surface vapor, and mixture partial vapor pressures, respectively. $R_{\mathbf{kin}}$ and $R_{\mathbf{dif}}$ are resistors related to each regime. $\dot{m}''$ is the mass flux of the gaseous carbon.

A simple particle agglomeration model is implemented based on spherical, monodisperse soot particles. Collision kernels are represented by two regimes: the free-molecular regime for low pressures and a continuum regime for higher pressures. The collision kernels are combined using a geometric average to obtain a combined coagulation coefficient.

E. Numerical Methods

The governing equations are solved using the HyBurn code developed at the University of Florida [13, 43]. The simulations use quad symmetry in 3D Cartesian coordinates. Symmetry and outflow boundary conditions are used at the lower and upper ends of the domain, respectively. A high-order Godunov approach is used for spatial discretization with fifth-order MUSCL [44, 45] and HLLC approximate Riemann solver [24, 46]. Third-order Runge-Kutta [25, 47] is used to advance the hydrodynamic terms in time. A CFL number of 0.5 is used until the return shock rebounds off the origin. The CFL number is then increased to 0.8 due to a reduction in solver difficulty. The VODE stiff ODE solver is used to integrate the chemical source terms [48]. The massively-parallel AMReX library [49] is used for adaptive mesh refinement with three levels of refinement, which corresponds to a 2048^3 grid at the finest level with a base grid of 256 x 256 x 256. The radiation is only solved on the base grid. The HyBurn code has been validated for an extensive list of test problems including detonations, flames, and aluminum particle dispersal [13, 24, 25, 43, 45, 50].

III. Results

The results show a post-detonation fireball with structures common to explosive blasts [4, 13, 43]. Figure 3 shows slice plots of gas temperature for Case 2 (120 mm) at various times. The radius and height are both scaled by f. At 500 μs, the primary shock can be seen propagating outwards. The return shock has stagnated and begins moving back towards the origin. The contact surface between the detonation products and air is well represented by high temperatures from exothermic chemical reactions.

By 1000 μs, the return shock has moved closer to the origin, entraining additional air into the fireball. This stretches the Rayleigh-Taylor instabilities into Rayleigh-Taylor "fingers". The "fingers" increase the surface area for reactions to occur, as shown by higher temperatures on their edges. At 2000 μs, the reshock is moving back through the fireball, locally increasing mixing. By 3000 μs, the primary shock has exited the fireball. Reactions continue to occur in the outer edge of the fireball where ambient oxygen meets the remaining detonation product gases.

Figure 4 shows slice plots of CO_2 mass fraction for Case 2 (120 mm) at various times. The elevated temperatures discussed in reference to Fig. 3 coincide with the elevated mass fractions of CO_2 in Fig. 4. At 500 μs, the mass fractions of CO_2 in the fireball are primarily at the initialized value as only reactions between detonation products have

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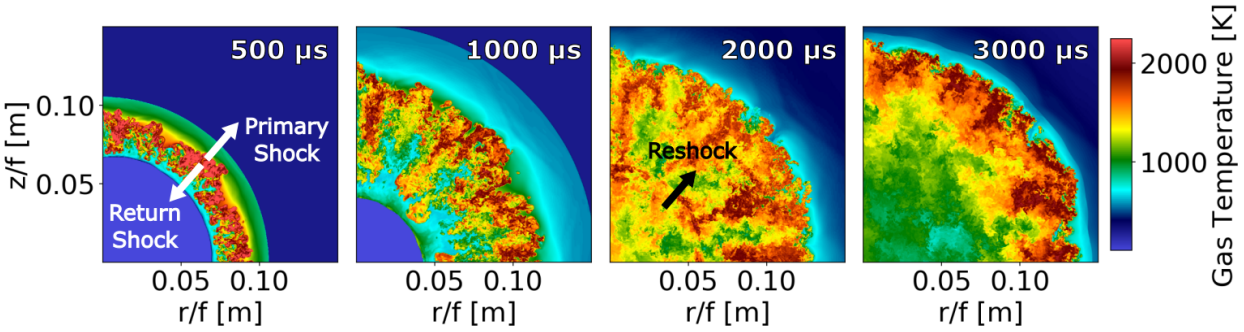


Fig. 3 Slice plots of gas temperature for Case 2 (120 mm) at 500, 1000, 2000, and 3000 μs.

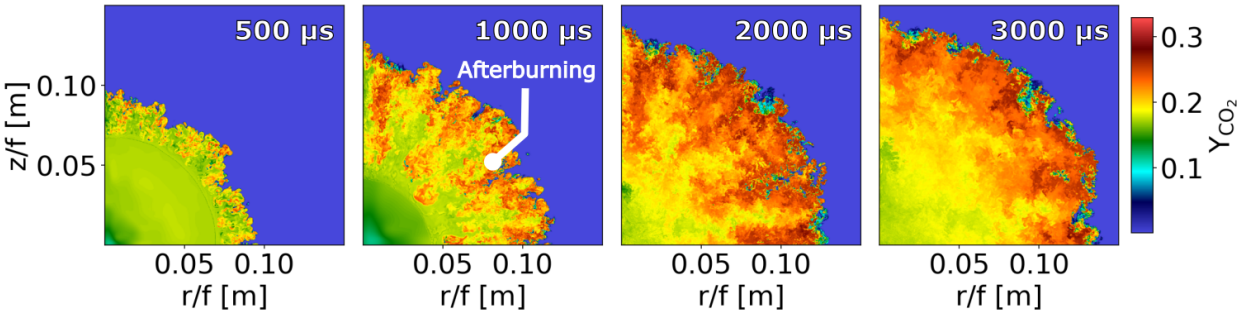


Fig. 4 Slice plots of CO_2 mass fraction for Case 2 (120 mm) at 500, 1000, 2000, and 3000 μs.

occurred. As time progresses, we see formation of CO_2 from the oxidation of CO and carbon soot particles. By 3000 μs, substantial afterburning has occurred.

Figure 5 shows a position-time diagram with the primary shock, return shock, and reshock. Time and space are scaled by f. A vacuum region is formed behind the return shock and is characterized by sub-ambient temperatures and pressures. Overall, the three simulations are gas-dynamically similar, save for slight differences in the speed of the reshocks. The initial conditions for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm) differ only by scaling size and domain with f. Prior studies have shown these flows to be gas-dynamically self-similar when combined with assumptions such as infinite-rate chemistry [8–12].

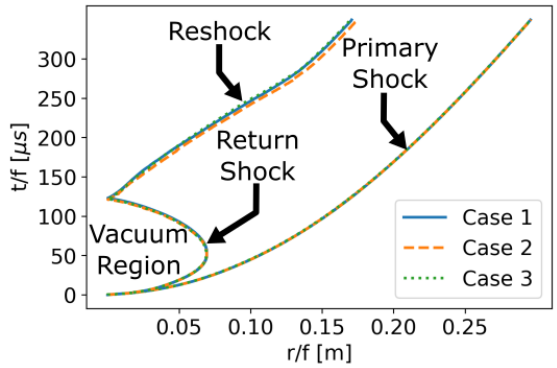


Fig. 5 Position-time diagram showing common fireball structures for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

The hemispherical charges are statistically symmetric. This geometry allows for azimuthal averaging of the fireball [1, 10]. Data is averaged based on the radial location of the cell center relative to the origin and weighted by cell volume. Figure 6 shows azimuthal averages of temperature and particle volume fraction for Cases 1, 2, and 3 at scaled 15 μs.

The early time flow is dominated by gas dynamics and all three cases behave almost identically. Chemical reactions and thermal particle radiation have not had time to make a noticeable impact on the flow.

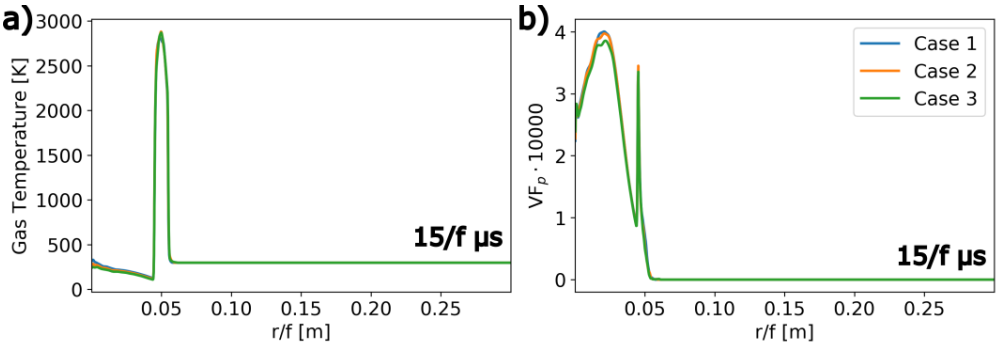


Fig. 6 Azimuthal averages of temperature and particle volume fraction for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm) at scaled 15 μs.

The heat release from exothermic reactions and heat loss from particle radiation begins to have an effect at later times. Figure 7 shows azimuthal averages of gas temperature for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm) at scaled 100 μs and 300 μs. The three cases no longer behave in a nearly identical manner, such as in Fig. 6. The larger charges show elevated temperatures in the fireball where the reactions between the detonation products and air are occurring.

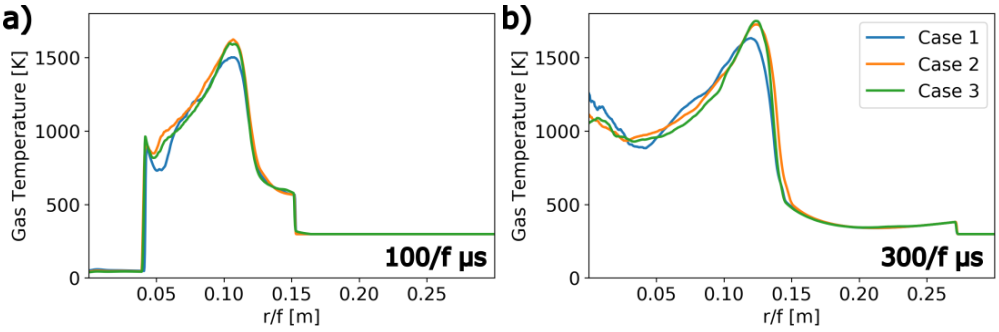


Fig. 7 Azimuthal averages of gas temperature for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm) at scaled 100 μs and 300 μs.

The afterburning shown in Fig. 4 is well exemplified by the position-time diagrams in Fig. 8. Figure 8 shows azimuthal averages of CO_2 mass fraction for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm). An ignition point can be seen where substantial reactions of oxygen with CO and carbon soot particles begin to occur. This ignition point coincides with shortly after the return shock reverses direction and begins to move back towards the origin. The entrainment of ambient air greatly increases the available oxidizer for the fuel-rich detonation products to burn.

IV. Discussion

A. Defining the Mixing Layer

The differences between the charge sizes centers around the behavior of the post-detonation flow where the detonation products mix and react with air. This is the mixing layer. The mixing layer is typically defined using a threshold method [51–53]. Boundaries between two different fluids are defined with thresholds such as 5% or 95% of some maximum value. The two fluids in this work are the detonation products and air. We define a mixture fraction using the Bilger definition [54] with a modification to include the carbon soot particles. The equations are

$$MF = \frac{\beta - \beta_{OX}}{\beta_{DP} - \beta_{OX}}, \tag{47}$$

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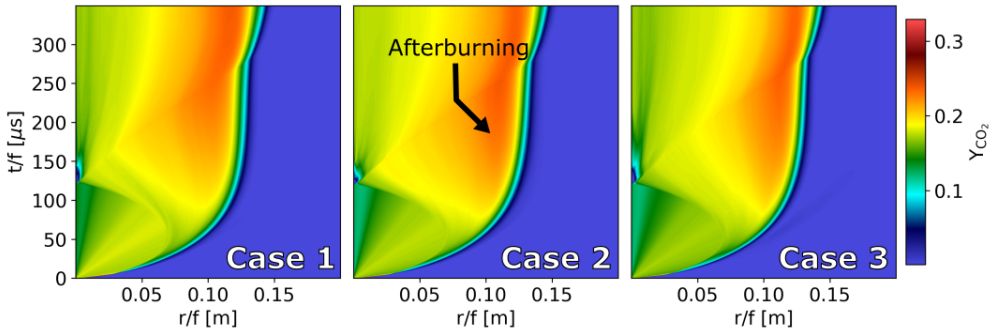


Fig. 8 Position-time diagram showing CO_2 mass fraction for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

$$\beta = C_1 \frac{Y_s}{MW_s} + Y_g \left(C_1 \frac{Z_C}{AW_C} + C_2 \frac{Z_H}{AW_H} + C_3 \frac{Z_O}{AW_O} \right), \tag{48}$$

$$Z_i = \sum_{j=1}^{N_g} \frac{a_{i,j} AW_i Y_j}{MW_j}, \tag{49}$$

where β_{OX} is the β value based on the ambient air composition and β_{DP} is the β value based on the detonation product composition at the CJ state. C_1, C_2, C_3 are the coefficients taken to be 2, 0.5, and -1. Y_s is the mass fraction of the carbon soot particles and Y_g is the mass fraction of the gas-phase. i is the element. Z_i is the mass fraction of element i. j is the species. N is the number of species. $a_{i,j}$ is the number of atoms of element i in species j. AW_i is the atomic weight of element i. Y_j is the mass fraction of species j. MW_j is the molecular weight of species j. MW_s is the molecular weight of soot. The mixture fraction is defined as 1 in the detonation products and 0 in the ambient air. We define the outer edge of the mixing layer where the mixture fraction is 0.05 and the inner edge of the mixing layer where the mixture fraction is 0.95.

The flow variables were averaged over mixture fraction, weighted by cell volume. Figure 9 shows mixture fraction averaged plots of temperature for Cases 1 (12 mm) and 3 (1200 mm) at various times. The black line indicates the averaged value and the background is colored by population sampled from slice plots. Mixture fraction increases with the x-axis, indicating a transition from the outer edge of the fireball to the inside of the fireball. Temperatures are initially high due to the energy released by the detonation. Expansion cools the fireball, but as time progresses the heat release from exothermic reactions maintains the temperature. The reshock moves through the fireball, as seen around scaled 200 μs, increasing mixing and reactions. Temperatures peak around a mixture fraction of 0.25. Case 3 exhibits a sharper peak that shifts towards a lower mixture fraction compared to Case 1.

Figure 10 shows mixture fraction averaged plots of CO_2 mass fraction for Cases 1 (12 mm) and 3 (1200 mm) at various times. Overall, both cases show the formation of CO_2 during afterburning processes. Additionally, the population clusters at higher mixture fractions at early time for both charge sizes, indicating that the fireball is mostly detonation products at this time. As time progresses, the population of cells shifts towards higher mixture fractions as mixing occurs. However, there are distinct differences between the charges. At scaled 50 μs, the CO_2 mass fraction for Case 1 within the fireball is relatively flat. Case 3 shows a peak at a mixture fraction around 0.25. This peak forms earlier than Case 1 and is indicative of afterburning.

Figure 11 shows mixture fraction-time diagrams of temperature for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm). The black dashed line represents an approximate location of the reshock. The white dashed line shows the stoichiometric mixture fraction. The unreacted detonation products have a stoichiometric mixture fraction of 0.233. The temperature is initially high following charge breakout, but cools as the blast expands. The reshock moves back through the fireball and locally increases temperature. It mixes the detonation products and air through the generation of impulsively driven Richtmyer-Meshkov instabilities. As charge size increases, the peak temperature in the mixing layer increases. In addition, the peak temperature shifts towards lower mixture fractions. More time for the detonation products and air to mix and react results in more heat release and more depletion of oxidizer. Therefore, higher temperatures occur and the reaction zone shifts closer to the ambient air.

Figure 12 shows mixture fraction-time diagrams of CO_2 mass fraction for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm). All three charge sizes show substantial afterburning in the form of CO_2 generation. The afterburning occurs

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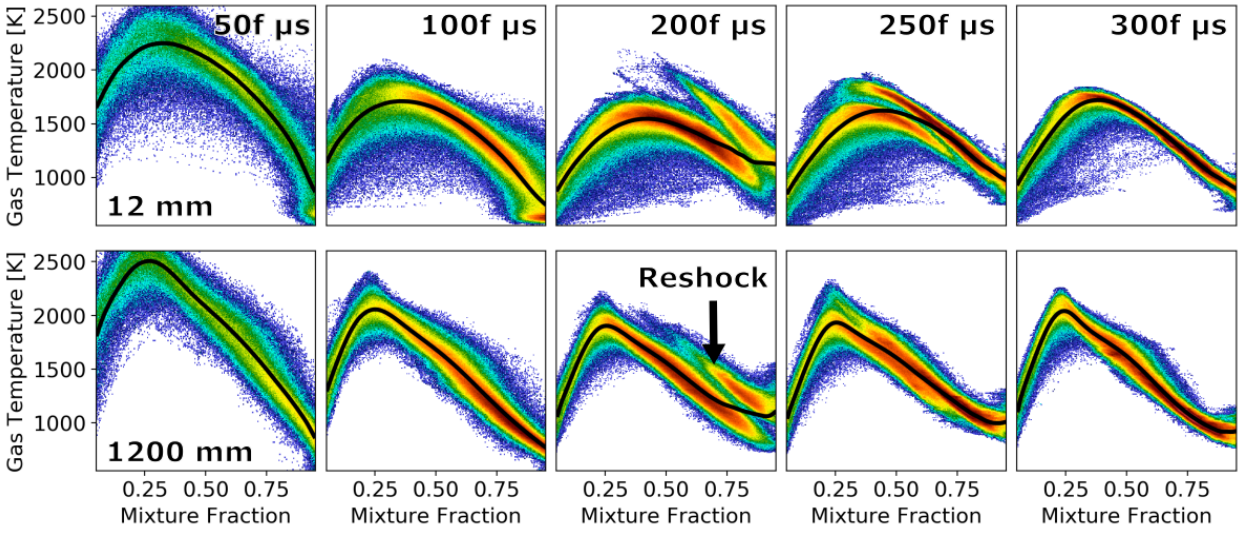


Fig. 9 Mixture fraction averaged plots of temperature for Cases 1 (12 mm) and 3 (1200 mm) at various times.

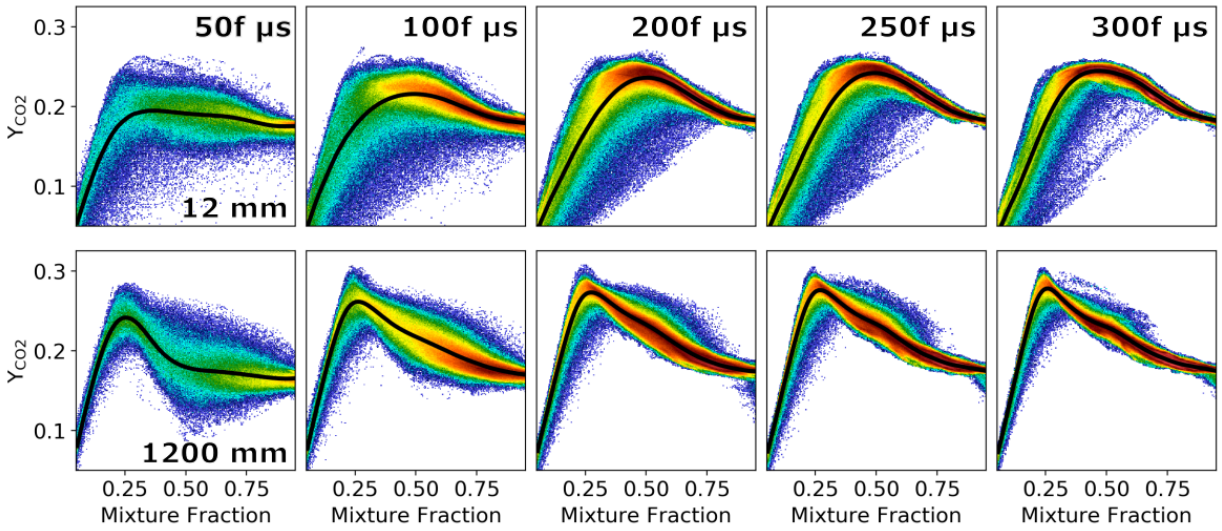


Fig. 10 Mixture fraction averaged plots of CO_2 mass fraction for Cases 1 (12 mm) and 3 (1200 mm) at various times.

comparatively earlier and closer to the ambient air in the larger charges. The peak CO_2 mixture fraction for the two largest charges is slightly fuel-rich of the stoichiometric mixture fraction. At later times, the profiles of the CO_2 mass fractions assumes a relatively linear relationship with mixture fraction. This suggests that a mixing-limited chemistry approach may be appropriate for TNT.

B. Impact of Finite-Rate Chemistry

In order to assess the importance of finite-rate chemistry to TNT fireballs, a Damköhler number is defined as $Da = \tau_{mix}/\tau_{chem}$, where τ_{mix} is a mixing timescale and τ_{chem} is a chemical timescale. Chemistry can be considered frozen for $Da \ll 1$. In this regime, the chemical timescales are slow compared to the mixing timescales such that the chemical composition shows little to no change. Chemistry is considered mixing-limited for $Da \gg 1$. In this regime, the reactions occur as fast as reactants can mix. Finite-rate chemistry is important between these two regimes. τ_{mix} is based on the integral timescale [55] and is calculated as

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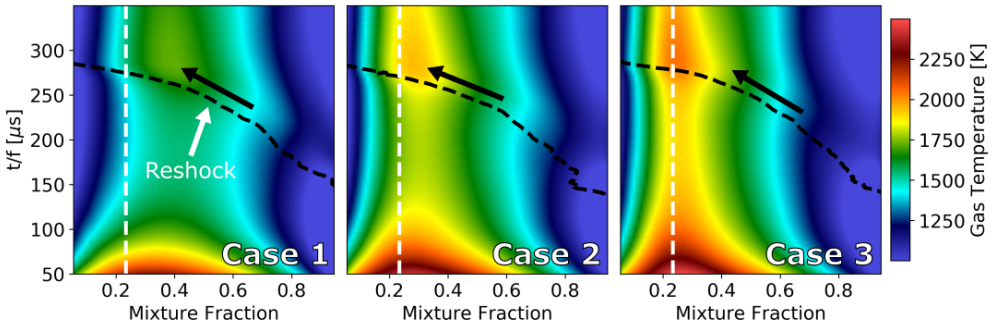


Fig. 11 Mixture fraction-time diagrams of temperature for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

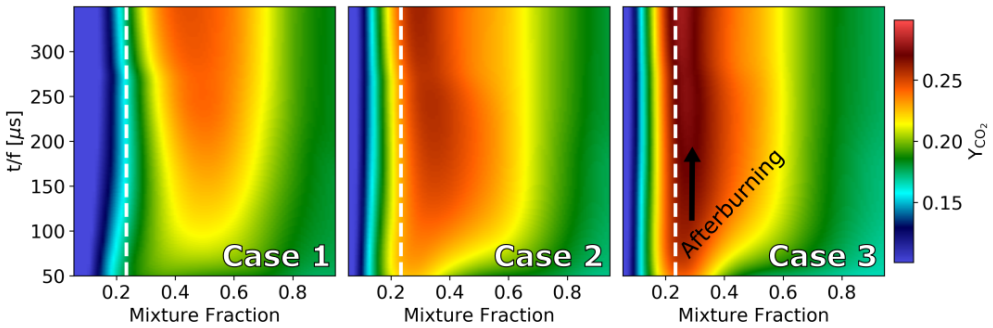


Fig. 12 Mixture fraction-time diagrams of CO_2 mass fraction for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

$$\tau_{\text{mix}} = \frac{k}{\epsilon}, \tag{50}$$

where k is the turbulent kinetic energy and ϵ is the TKE dissipation. The TKE includes all three coordinates as

$$k = \frac{1}{2}\sum_{i}^{N} \overline{u_i' u_i'}, \tag{51}$$

where i is the coordinate (r, θ, and ϕ) and N is the total number of coordinates. The velocity fluctuations are approximated using the azimuthal averages with the relation

$$\overline{u_i' u_i'} = \overline{u_i^2} - \overline{u_i}^2. \tag{52}$$

The TKE dissipation rate [56, 57] is approximated with

$$\epsilon \approx \frac{1}{\ell}\sum_{i}^{N} \hat{u}_i^3, \tag{53}$$

where ℓ is the integral length scale. The integral length scale is taken to be the mixing layer width, which we define as $r_{0.05} - r_{0.95}$. This is the difference in radial location of the outer and inner boundaries of the mixing layer. $\hat{u_i}$ is the turbulence intensity calculated as

$$\hat{u}_i = \sqrt{\overline{u_i' u_i'}}. \tag{54}$$

τ_{chem} is often defined by selecting pertinent species and using their production and depletion rates [58, 59], but we find that the chemical energy heat release rate does a good job of representing the averaged chemistry. Therefore, we base τ_{chem} on the chemical energy heat release rate of both the gas- and particle-phase. It is calculated as

$$\tau_{\text{chem}} = \frac{\alpha_{\text{g}}\rho_{\text{g}}h_{\text{g}} + \alpha_{\text{s}}\rho_{\text{s}}h_{\text{s}}}{\dot{Q}_{\text{g}} + \dot{Q}_{\text{s}}}, \tag{55}$$

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where h_g and h_s are the absolute values of gas- and particle enthalpies, respectively. $\dot{Q}_g$ and $\dot{Q}_s$ are the chemical heat release rates of the gas- and particle-phases, respectively. Overall, the gas-phase contribution dominates the particle-phase contribution.

Figure 13 shows position-time diagrams of Damköhler number for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm). The vacuum region and ambient have been removed as the chemical and mixing timescales in those regions are not relevant. The black line indicates a Damköhler number of 0.1 while the red line indicates a Damköhler number of 10. The mixing layers in all three cases exist in the mixing-limited regime where the bulk of the afterburning reactions occur. The region where finite-rate chemistry is important is relatively small. This result gives support to the applicability of mixing-limited chemistry to the post-detonation afterburning of explosives such as TNT with highly negative oxygen balances.

Another interesting feature of Fig. 13 is the frozen chemistry region behind the reshock. This region grows with increasing charge size. More oxidizer is depleted within the fireball in the larger charges, which shifts the afterburning region closer towards the ambient air interface.

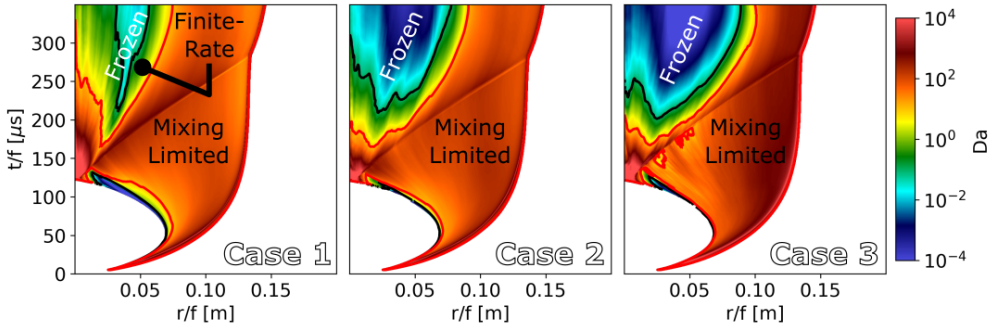


Fig. 13 Position-time diagrams of Damköhler number for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

C. Influence of Particle Radiation

Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm) included particle radiation. Figure 14 shows position-time diagrams of incident radiation for Cases 1, 2, and 3. The black dashed lines show the boundaries of the mixing layer. All three cases show strong incident radiation when the primary shock breaks out of the charge and the detonation products begin to mix with the shock heated air. All three cases also show strong incident radiation during the formation of the reshock. The incident radiation from the reshock formation propagates out of the fireball in Case 1, as shown in Fig. 14. The larger volume of particles in the larger charge sizes results in more absorption, neutralizing this effect. This implies that the smallest case is more optically thin than the larger cases.

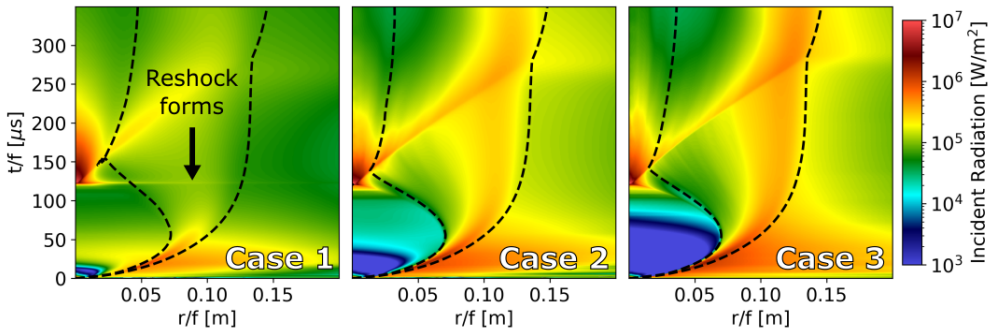


Fig. 14 Position-time diagrams of incident radiation for Cases 1 (12 mm), 2 (120 mm), and 3 (1200 mm).

In order to assess the impact of thermal radiation, we compare Cases 1 (12 mm) and 3 (1200 mm) with 4 (12 mm) and 5 (1200 mm). Cases 4 and 5 are identical to 1 and 3, but do not solve the RTE. Figure 15 shows azimuthal averages of gas temperature and CO_2 mass fraction for Cases 1, 3, 4, and 5. The difference between the two 12 mm cases is

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negligible. The thermal radiation time scales are too slow compared to the flow and chemical timescales to have a noticeable impact. However, the two 1200 mm cases show minor differences. The peak gas temperature is higher in the non-radiative case by about 5-6%. Additionally, the mass fraction of CO_2 is slightly lower in the non- case at late time. This difference may stem from the additional CO produced by soot reactions at the higher temperatures.

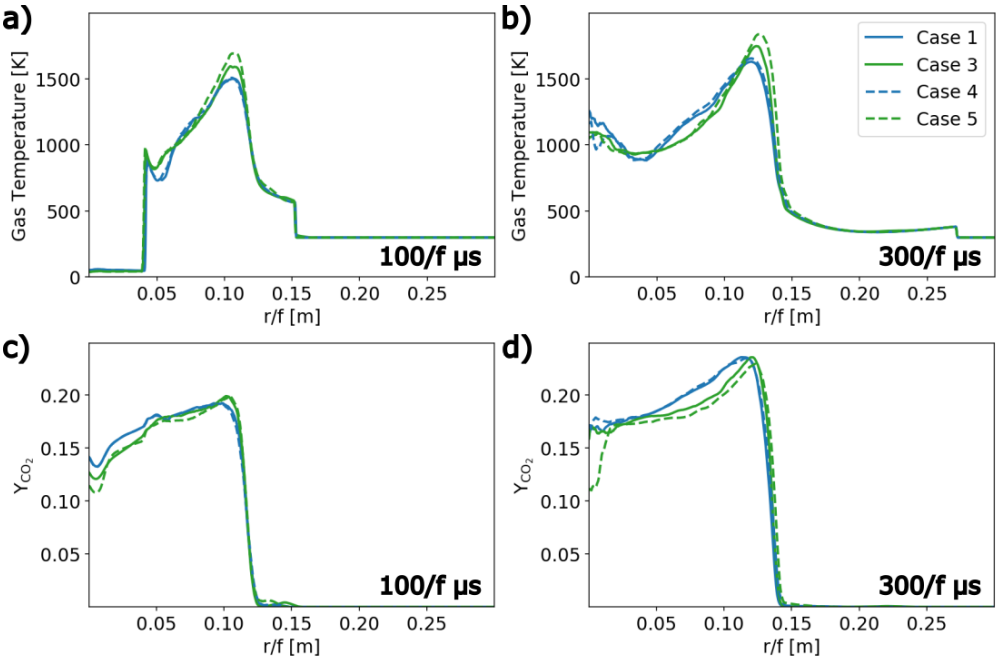


Fig. 15 Azimuthal averages of gas temperature and CO_2 mass fractions for Cases 1 (12 mm), 3 (1200 mm), 4 (12 mm no radiation), and 5 (1200 mm no radiation) at scaled 100 μs and 300 μs.

V. Conclusion

Numerical simulations of post-detonation TNT afterburning were performed in 3D Cartesian coordinates. The simulations considered finite-rate detailed chemistry with a skeletal chemical reaction mechanism consisting of 43 species and 268 reactions. The simulation also considered the behavior of carbon soot particles with thermal radiation. The results show structures common to explosive blasts.

The chosen geometry allows for azimuthal averaging of the flow. Early time behavior is gas dynamically driven and scales with charge size. Later time behavior is shown to be influenced by chemical reactions. Higher temperatures are found in the larger charge sizes. All charge sizes show strong afterburning through the generation of CO_2.

The calculation of a Damköhler number showed that the mixing layers in all three charge sizes exist in the mixing-limited chemical regime. This result suggests that the mixing-limited chemistry approach common to explosive blasts may be appropriate for explosives with fuel-rich detonation products like TNT.

Thermal radiation was shown to have a negligible effect on the 12 mm charges. The azimuthal averaged temperatures and mass fractions of CO_2 showed little to no difference between cases with and without radiation. Thermal radiation was shown to have a larger impact on the 1200 mm charge. The longer timescales in the larger charge allowed more time for the particles to radiate. This resulted in 5-6% higher peak temperatures at 30000 μs in the non-radiative case. The current simulations did not consider absorption and emission of the gas-phase. The shock heated air following charge breakout reaches above 12,000 K. Recombination of dissociated air generates substantial NO and NO_x species with non-negligible cross sections. Future work should include gas-phase emission and absorption to better explore the importance of thermal radiation in high explosive blasts.

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